Remark 1.2. If S is simply connected, equation (6) implies in particular that $\det(\nabla^2 v) = 0$. For a proof of this result see [15, Proposition 9].

The case $\alpha = 4$ corresponds to the Von Kármán model, that is,

$$E_j^{\mathrm{VK}}(u,v) = \frac{1}{24}\int_S \bar{Q}_j(\nabla^2 v)\,dx + \frac{1}{8}\int_S \bar{Q}_j(\nabla u^T + \nabla u + |U_j^{-1}e_3|^2 \nabla v \otimes \nabla v)\,dx\,,$$

for $v \in W^{2,2}(S)$ and $u \in W^{1,2}(S;\mathbb{R}^2)$.

Lastly, if $\alpha > 4$, we prove the Γ-convergence to the linearized Von Kármán model:

$$E_j^{\mathrm{LVK}}(u,v) = \frac{1}{24}\int_S \bar{Q}_j(\nabla^2 v)\,dx + \frac{1}{8}\int_S \bar{Q}_j(\nabla u^T + \nabla u)\,dx\,,$$

for $v \in W^{2,2}(S)$ and $u \in W^{1,2}(S;\mathbb{R}^2)$. The following theorem summarizes these Γ-convergence results.

Theorem 1.3 (Γ-convergence ($\alpha > 2$))**.** *Suppose $\alpha > 2$.*

(i) For any sequence $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ that satisfies $E_h^\alpha(y_h) \le C$ for every $h > 0$ there exist an index $j \in \{1,\dots,l\}$, two sequences $(\bar{R}_h) \subset \mathrm{SO}(3)$, $(c_h) \subset \mathbb{R}$, and two maps $v \in W^{2,2}(S)\,, u \in W^{1,2}(S;\mathbb{R}^2)$ such that, up to a subsequence, the following convergences hold:

(a) $u_h \rightharpoonup u$ in $W^{1,2}(S;\mathbb{R}^2)$,

(b) $v_h \to v$ in $W^{1,2}(S)$,

where u_h and v_h are, respectively, the in-plane and out-of-plane displacements around the well K_j defined as in (4)–(5) for the roto-translated deformation

$$\tilde{y}_h = \bar{R}_h y_h + c_h\,.$$

Moreover, if $2 < \alpha < 4$, then

$$\nabla u^T + \nabla u + |U_j^{-1}e_3|^2 \nabla v \otimes \nabla v = 0\,. \tag{7}$$

(ii) For any sequence $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ as in (i)

(a) if $2 < \alpha < 4$, then $\liminf_{h\to 0} E_h^\alpha(y_h) \ge E_j^{\mathrm{CVK}}(v)$,

(b) if $\alpha = 4$, then $\liminf_{h\to 0} E_h^\alpha(y_h) \ge E_j^{\mathrm{VK}}(u,v)$,

(c) if $\alpha > 4$, then $\liminf_{h\to 0} E_h^\alpha(y_h) \ge E_j^{\mathrm{LVK}}(u,v)$.

(iii) Suppose that $2 < \alpha < 4$ and that S is simply connected and satisfies (2). For any choice of $j \in \{1,\dots,l\}$ and $v \in W^{2,2}(S)$ such that there exists $u \in W^{1,2}(S;\mathbb{R}^2)$ solving (7), there exists a sequence $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ such that

(a) $v_h \to v$ in $W^{1,2}(S)$, where v_h is defined as in (5),

(b) $\lim_{h\to 0} E_h^\alpha(y_h) = E^{\mathrm{CVK}}(v)$.

(iv) *For any choice of* $j \in \{1,\dots,l\}$, $u \in W^{1,2}(S;\mathbb{R}^2)$, *and* $v \in W^{2,2}(S)$ *there exists a sequence* $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ *such that, defining* $u_h\,, v_h$ *as in* (4)-(5),

(a) $v_h \to v$ *in* $W^{1,2}(S)$,

(b) $u_h \rightharpoonup u$ *in* $W^{1,2}(S;\mathbb{R}^2)$,

(c) $\lim_{h\to 0} E_h^\alpha(y_h) = E^{\mathrm{VK}}(u,v)$ *if* $\alpha = 4$,

(d) $\lim_{h\to 0} E_h^\alpha(y_h) = E^{\mathrm{LVK}}(u,v)$ *if* $\alpha > 4$.

Remark 1.4. In the proof of Theorem 1.3–(iii) we cannot use the truncation argument of [16, 15]. Indeed, the penalty term in the energy requires higher regularity. To overcome this issue we suppose that S satisfies (2), so that we can apply the approximation result given in [17].

1.2.3 Convergence of minimizers in the presence of dead loads

External forces can be included in the previous analysis. We suppose $q > 1$, and we study the convergence of minimizers of the rescaled total energy

$$J_h^\alpha \colon W^{1,2}(\Omega;\mathbb{R}^3) \to \mathbb{R} \cup \{+\infty\}\,, \quad J_h^\alpha(y_h) = E_h^\alpha(y_h) - \frac{1}{h^\alpha}\int_\Omega f_h \cdot y_h\, dx\,,$$

where $f_h \colon S \to \mathbb{R}^3$ is a sequence of dead loads that satisfies

$$\frac{1}{h^{\gamma+1}} f_h \to f \quad \text{in } L^{q'}(S;\mathbb{R}^3)\,. \tag{8}$$

Here q' is the conjugate exponent of q. Note that $q' \geq 2$, so that the strong convergence of the forces holds also in $L^2(S;\mathbb{R}^3)$. We assume the forces to be mean-free, i.e.,

$$\int_S f_h\, dx = 0\,,$$

otherwise the infimum of J_h^α is $-\infty$. We prove the following result.

Theorem 1.5. *Suppose that* S *satisfies* (2) *and* $q > 1$. *Suppose that* $(y_h) \subset W^{1,2}(\Omega;\mathbb{R}^3)$ *is a sequence of deformations that are quasi-minimizers for* J_h^2, *i.e.,*

$$\limsup_{h\to 0}\,(J_h^2(y_h) - \inf J_h^2) = 0\,.$$

Then, $E_h^2(y_h) \leq C$ *for every* $h > 0$ *and there exist* $y \in W^{2,2}(S;\mathbb{R}^3)$ *and* $j \in \{1,\dots,l\}$ *satisfying* $(\nabla y)^T \nabla y = (U_j^2)'$ *such that, up to subsequences,* $y_h \to y$ *in* $W^{1,2}(\Omega;\mathbb{R}^3)$ *and* (j,y) *minimizes*

$$J_j^{\mathrm{K}}(y) = E_j^{\mathrm{K}}(y) - \int_S f \cdot y\, dx\,,$$

over the set

$$\{(j,y) \in \{1,\dots,l\} \times W^{2,2}(S;\mathbb{R}^3) \colon \nabla y^T \nabla y = (U_j^2)'\}\,.$$

For the case $\alpha > 2$ we need to introduce some notation. We denote by $\mathcal{M}_h \subset K$ the set of maximizers of the functional

$$F_h \colon K \to \mathbb{R}, \quad RU_j \mapsto \int_S f_h \cdot RU_j \begin{pmatrix} x' \\ 0 \end{pmatrix} dx\,.$$

Note that $\mathcal{M}_h$ is not empty by compactness of K. Similarly, we define

$$\mathcal{M} = \operatorname*{argmax}_K F\,,$$

where

$$F(A) = \int_S f \cdot A \begin{pmatrix} x' \\ 0 \end{pmatrix} dx\,.$$

We define the sets $\mathcal{R}^j$ and $\mathcal{R}^j_h$ as follows

$$\begin{aligned} \mathcal{R}^j &= \operatorname*{argmax}_{R \in \mathrm{SO}(3)} F(RU_j)\,, \\ \mathcal{R}^j_h &= \operatorname*{argmax}_{R \in \mathrm{SO}(3)} F_h(RU_j)\,. \end{aligned}$$

By definition of K, there are subsets of indices $\Lambda_h, \Lambda \subset \{1, \dots, l\}$ such that $\mathcal{M}_h = \bigcup_{j \in \Lambda_h} \mathcal{R}^j_h U_j$ and $\mathcal{M} = \bigcup_{j \in \Lambda} \mathcal{R}^j U_j$. These definitions generalize to our context the notion of optimal rotations introduced in [21]. One can prove that $\mathcal{R}^j_h$ and $\mathcal{R}^j$ are closed, connected, boundaryless, and totally geodesic submanifolds of SO(3). Indeed, the proof in [21] does not rely on the specific structure of F or F_h, but only on their linearity. We denote by $T\mathcal{R}^j_R$ and $N\mathcal{R}^j_R$ the tangent space and the normal space to $\mathcal{R}^j$ at the point R, respectively. By [21, Proposition 4.1], we have

$$\begin{aligned} T\mathcal{R}^j_R &= \{RW \in \mathbb{R}^{3\times 3} \colon W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}\,, F(RW^2 U_j) = 0\}\,, \\ N\mathcal{R}^j_R &= \{RW \in \mathbb{R}^{3\times 3} \colon W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}\,, RW \perp T\mathcal{R}^j_R\}\,. \end{aligned}$$

Similarly, we define $T\mathcal{R}^j_{hR}$ and $N\mathcal{R}^j_{hR}$. We can define the projection operators P^j and P^h_j of SO(3) onto $\mathcal{R}^j$ and $\mathcal{R}^j_h$, respectively. These projections have to be understood with respect to the intrinsic distance of SO(3), i.e.,

$$\mathrm{dist}_{\mathrm{SO}(3)}(R, Q) = \min\left\{|W| \colon W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}\,, Q = Re^W\right\}\,,$$

and are well-defined at least in a neighborhood of $\mathcal{R}^j$ and $\mathcal{R}^j_h$, respectively. We will suppose that the forces f_h are such that

(F1) $\Lambda_h = \Lambda$ for $h \ll 1$,

(F2) $\dim \mathcal{R}^j_h \to \dim \mathcal{R}^j$ for any $j \in \Lambda$.

Note that in general one only has $\Lambda_h \subseteq \Lambda$ and $\limsup_{h\to 0} \dim R^j_h \le \dim R^j_h$, as shown in the following example. The failure of (F1)–(F2) may happen, for instance, when the direction along which the force acts is slightly perturbed.

Example 1.6. Let $\alpha > 2$ and set $S = (-\frac{1}{2}, \frac{1}{2})^2$. Suppose that $l = 2$ and let

$$U_1 = \mathrm{Id}, \qquad U_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$